THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The area of the triangle formed by $(7, 0)$, $(0, 9)$ and the origin, doubled, is: 1

(a) 63 **(b)** 16 **(c)** $\frac{63}{2}$ **(d)** 126

2. If $\tan\theta = \frac{4}{3}$, then $\dfrac{1-\tan^2\theta}{1+\tan^2\theta} =$ 1

(a) $\frac{24}{25}$ **(b)** $\frac{3}{4}$ **(c)** $-\frac{7}{25}$ **(d)** $\frac{7}{25}$

3. A flagstaff stands on the top of a building 10 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 30° and 60° respectively. The length of the flagstaff is: 1

(a) 20 m **(b)** $20\sqrt{3}$ m **(c)** 30 m **(d)** 40 m

4. PA and PB are tangents from P to a circle, each of length 7 cm. A third tangent touches the circle at E on the minor arc AB and meets PA at C and PB at D. The perimeter of $\triangle PCD$ is: 1

(a) 21 cm **(b)** 14 cm **(c)** 7 cm **(d)** 9 cm

5. A brooch is a circular piece of silver wire of diameter 21 mm. The wire is also used for 4 equal diameters that divide the brooch into 8 sectors. The total length of wire required (in mm) is: 1

(a) 66 mm **(b)** 108 mm **(c)** 346.5 mm **(d)** 150 mm

6. A solid sphere of radius 10.5 cm fits exactly inside a cylinder (it touches the curved surface and both bases). The volume of the space in the cylinder not occupied by the sphere is: 1

(a) 7276.5 cm³ **(b)** 2425.5 cm³ **(c)** 4851 cm³ **(d)** 12127.5 cm³

7. The median of a distribution with classes 200–210, 210–220, 220–230, 230–240, 240–250, 250–260, 260–270 and frequencies 12, 8, x, 25, y, 18, 19 respectively, and total frequency 100, is 238.4. The value of x is: 1

(a) 10 **(b)** 9 **(c)** 8 **(d)** 11

8. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the sum is less than 5 is: 1

(a) $\frac{5}{6}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{5}{36}$ **(d)** $\frac{7}{36}$

9. The largest number dividing 67, 103 and 175 leaving remainder 4 in each case is: 1

(a) 18 **(b)** 1 **(c)** 3 **(d)** 9

10. If α, β are zeroes of $x^2 - 11x + 18$, then $|\alpha - \beta|$ is: 1

(a) 11 **(b)** 8 **(c)** 7 **(d)** 14

11. If $11x + 4y = 53$ and $5x + 3y = 30$, then $x + y =$ 1

(a) 15 **(b)** 5 **(c)** 3 **(d)** 8

12. For which value of k does $x^2 + kx + 9 = 0$ have equal roots (take $k > 0$)? 1

(a) 12 **(b)** 6 **(c)** 9 **(d)** 3

13. In an AP, $S_8 = 200$ and $S_7 = 154$. Then $a_8 =$ 1

(a) 46 **(b)** 52 **(c)** 40 **(d)** 354

14. The altitude on the hypotenuse of the right triangle with legs 3 and 4 is of length (cm): 1

(a) $\frac{7}{2}$ **(b)** $\frac{5}{2}$ **(c)** 6 **(d)** $\frac{12}{5}$

15. If $P(x, y)$ is equidistant from $(1, 3)$ and $(7, 5)$ and lies on $y = 0$, then $x =$ 1

(a) $\frac{32}{3}$ **(b)** 4 **(c)** $\frac{16}{3}$ **(d)** $-\frac{16}{3}$

16. If $\sin\theta = \frac{4}{5}$, the value of $2\sin\theta\cos\theta$ is: 1

(a) $\frac{8}{5}$ **(b)** $\frac{6}{5}$ **(c)** $\frac{24}{25}$ **(d)** $\frac{12}{25}$

17. A car moves at a uniform speed towards the foot of a vertical tower. The angle of elevation of the top of the tower changes from 30° to 60° in 4 minutes. The further time the car takes to reach the foot of the tower is: 1

(a) 4 minutes **(b)** 2 minutes **(c)** $\frac{4}{3}$ minutes **(d)** 8 minutes

18. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (15x + 4)$ cm and $QB = (8x + 25)$ cm. The length of QA is: 1

(a) 51 cm **(b)** 49 cm **(c)** 70 cm **(d)** 3 cm

19. A horse is tied at the midpoint of one side of a large square field with a rope 35 m long. The area of the field it can graze is: 1

(a) 1925 m² **(b)** 3850 m² **(c)** 55 m² **(d)** 110 m²

20. A solid cylinder of radius 6 cm and height 5 cm is melted and recast into a cone of the same base radius. The height of the cone is: 1

(a) 10 cm **(b)** 15 cm **(c)** 5 cm **(d)** 30 cm

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Show that the points $A(0, 0)$, $B(3, 4)$, $C(8, 4)$, $D(5, 0)$ are the vertices of a rhombus. 2

22. If $\sin\theta = \frac{4}{5}$, find the value of $\cos^2\theta - \sin^2\theta$. 2

23. The angle of elevation of the top of a vertical tower from a point on level ground is 45°. After walking 6 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower. 2

24. Tangents PA and PB are drawn from a point P, 75 cm from the centre O of a circle of radius 60 cm. The length of each tangent is 45 cm. Find the length of the chord AB. 2

25. An umbrella has 14 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 49 cm. Find the area between two consecutive ribs. 2

26. A solid sphere of radius 6 cm is completely dropped into water in a cylindrical vessel of base radius 8 cm, and no water spills. Find the rise in the water level. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. A building is $18\sqrt{3}$ m high. From the top of the building the angle of elevation of the top of a tower is 30° and from the foot of the building it is 60°. Find the height of the tower and its distance from the building (in this order). 3

28. Tangents PA and PB are drawn from a point P, 50 cm from the centre O of a circle of radius 30 cm. The length of each tangent is 40 cm. Find the length of the chord AB. 3

29. A sector of a circle of radius 42 cm has central angle 50°. Find the area of the sector. 3

30. A solid metallic cylinder of base radius 9 cm and height 32 cm is melted and recast into solid spheres of radius 6 cm. Find the number of spheres obtained. 3

31. The marks obtained by students in a test are given below. 3

Marks	40–50	50–60	60–70	70–80	80–90	90–100	100–110	110–120
Number of students	11	19	8	20	23	2	13	4

Find the mean and the mode (in this order).

OR

Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that the sum is 9. 3

32. Find the greatest number that divides 1251 and 9377 leaving remainder 1 in each case. 3

OR

The zeroes of $x^2 - 10x + c$ are α and β with $\alpha - \beta = 4$. Find c. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Prove that if PA and PB are tangents from P and a third tangent touches the circle at C on the minor arc AB, meeting PA at X and PB at Y, then $XY = AX + BY$ and the perimeter of $\triangle PXY$ is $2PA$. 5

34. A chord of a circle of radius 35 cm subtends an angle of 60° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) Find the area of the minor segment. 5

35. A toy is a cone of base radius 49 cm and height 168 cm mounted on a hemisphere of the same radius (the flat faces coincide). Find the total surface area of the toy. 5

36. The median of a distribution with classes 200–210, 210–220, 220–230, 230–240, 240–250, 250–260, 260–270 and frequencies 12, 8, x, 25, y, 18, 19 respectively, and total frequency 100, is 238.4. Find the value of x. 5

OR

Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) the product of the numbers is 12,
(ii) the product is a perfect square,
(iii) the first number is greater than the second.

37. Prove that $\dfrac{1}{\sqrt{2}}$ is irrational and hence that $\dfrac{3}{\sqrt{2}}$ is irrational. 5

OR

If α, β are the zeroes of $x^2 - 3x - 10$, form a quadratic polynomial whose zeroes are $\alpha + \beta$ and $\alpha\beta$. 5

38. A man travels 360 km partly by train and partly by bus. If he travels 150 km by train and the rest by bus, he takes $\frac{41}{4}$ hours; if he travels 240 km by train and the rest by bus, he takes 11 hours. Find the speeds of the train and the bus. 5

OR

A motorboat goes 48 km upstream and 48 km downstream in 7 hours. If the speed of the stream is 2 km/h, find the speed of the boat in still water. 5

END OF QUESTION PAPER **Total: 80 Marks**

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